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الجامعة العبرية في القدس

ON THE LANDSCAPE OF SPOKEN LANGUAGE MODELS

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AGENDA

- Speech language models
 - Motivation
 - Categorization and definitions
- Progress and scaling laws
- Going beyond spoken context in SLM evaluation
- Discussion & future directions



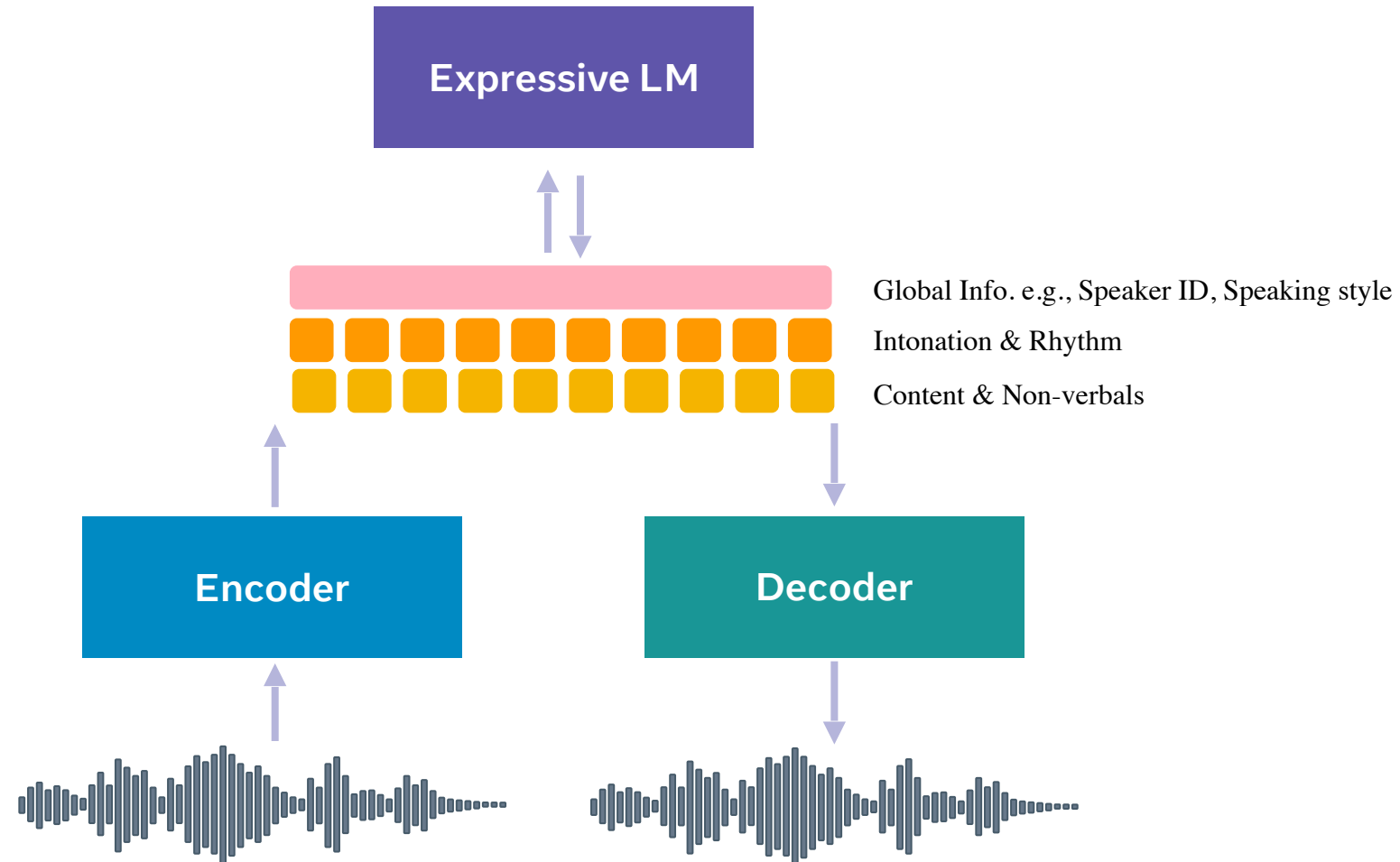


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SPEECH LANGUAGE MODELS

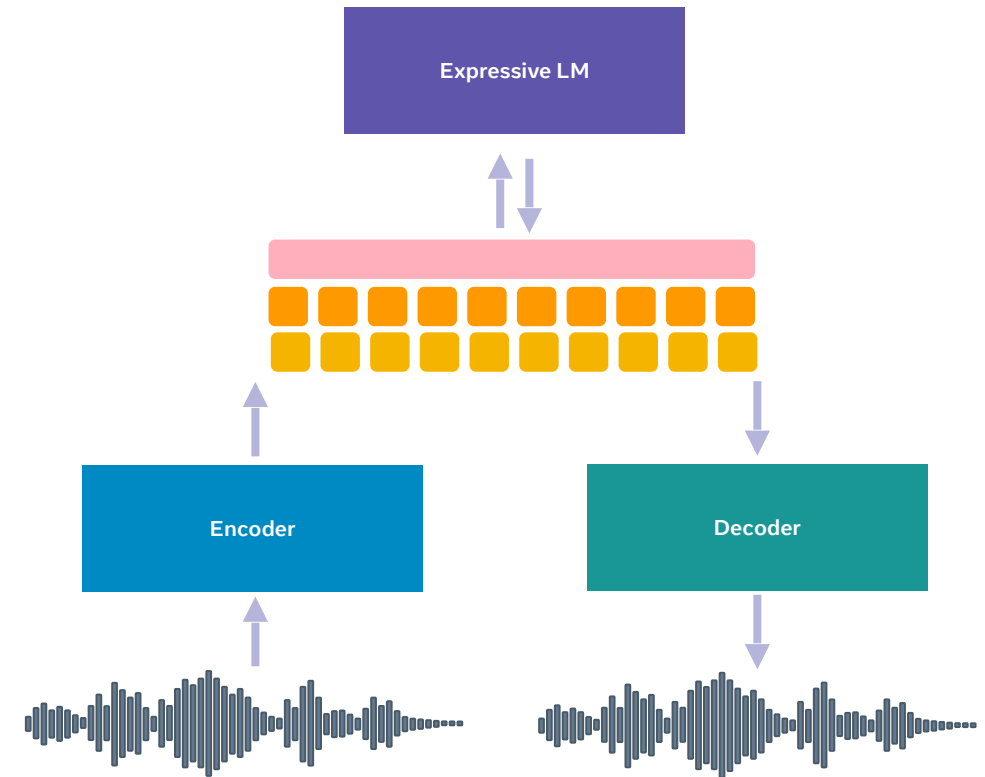


SPEECH LANGUAGE MODELS



SPEECH LANGUAGE MODELS

- Different types of SLMs
 - e.g., textless-SLMs
 - Joint speech-text SLMs
 - Speech-aware LMs



- For more info about SLMs:

Arora, Siddhant, et al. "On the landscape of spoken language models: A comprehensive survey."



SPEECH LANGUAGE MODELS

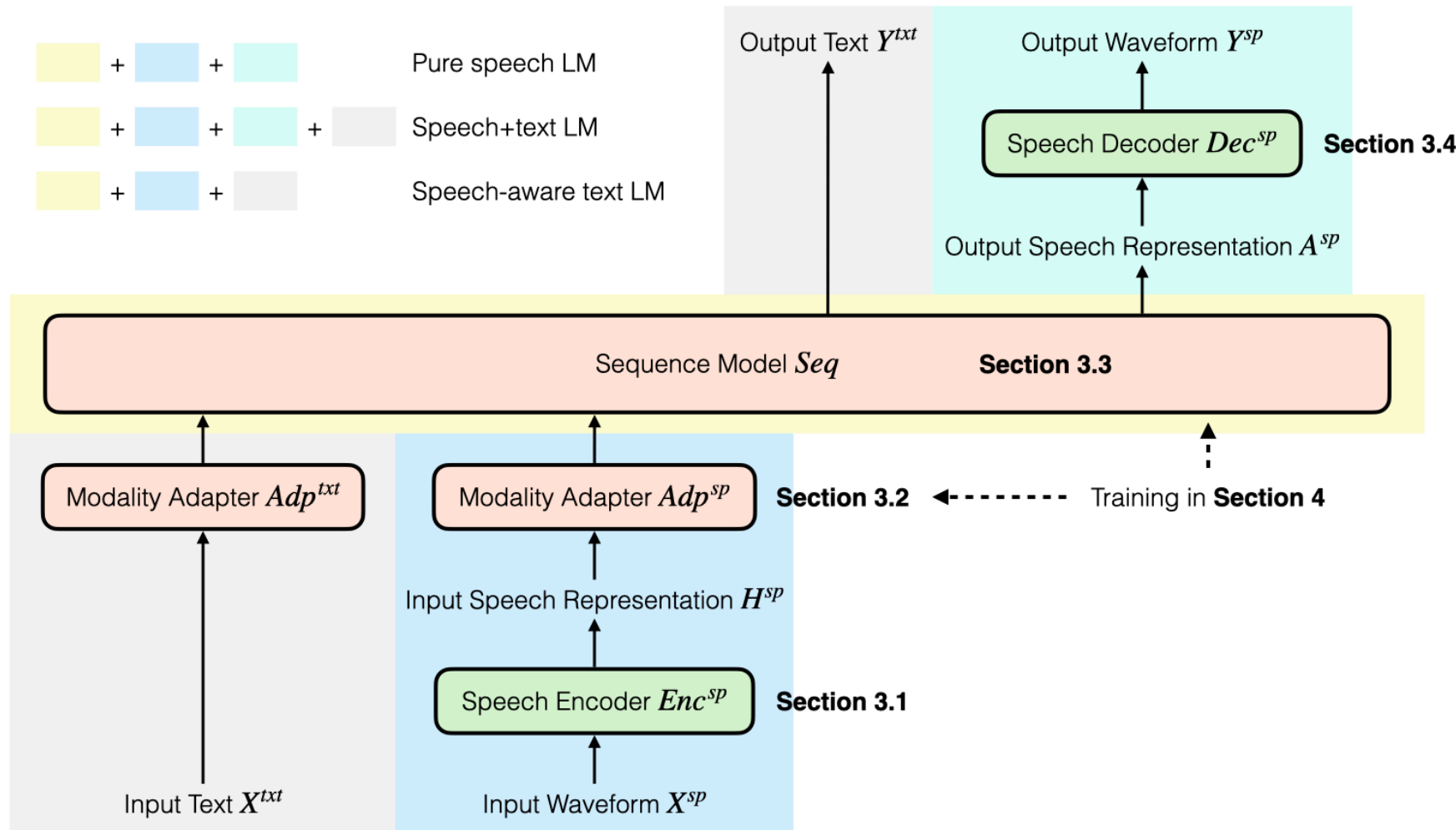
Type of LM	Training Strat.	Distribution	Example
TextLM	pre-training	$p(\text{text})$	GPT, Llama
TextLM	post-training	$p(\text{text} \mid \text{text})$	ChatGPT, Llama-Inst.
textless-SpeechLM	pre-training	$p(\text{speech})$	GSLM, AudioLM
textless-SpeechLM	post-training	$p(\text{speech} \mid \text{speech})$	AlignSLM, Slamming
Joint speech-text LM	pre-training	$p(\text{text}, \text{speech})$	SpiRitLM, Moshi
Joint speech-text LM	post-training	$p(\text{text}, \text{speech} \mid \text{text}, \text{speech})$	Moshi, Mini-Omni
Speech-aware LM	post-training	$p(\text{text} \mid \text{speech}, \text{text})$	Salmonn, Qwen-Audio-Chat



SPEECH LANGUAGE MODELS

- *Universal Speech Processing Systems*
- Definition:
 - It has both **spoken input** and **spoken output** with optional text input and/or output. The spoken input may serve as either an instruction or a context.
 - It is intended to be “**universal**”; that is, it should in principle be able to address **arbitrary spoken language tasks**, including both traditional tasks and more complex reasoning about spoken data.
 - It takes **instructions** or **prompts** in the form of natural language (either speech or text), and not, for example, task specifiers or soft prompts.

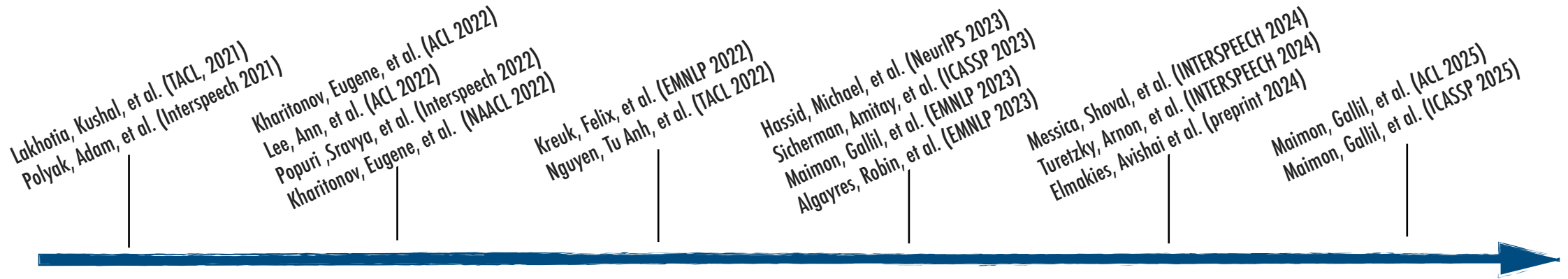
SPEECH LANGUAGE MODELS





TEXTLESS-SLMs

- In textless-SLMs we consider speech-only tokens (no textual tokens)
- We've made a lot of progress!



- Current textless-SLMs can be consistent with:
 - Speaker id, acoustics, etc.
 - Syntax is mostly ok
 - Semantics is not good :(

EXAMPLE



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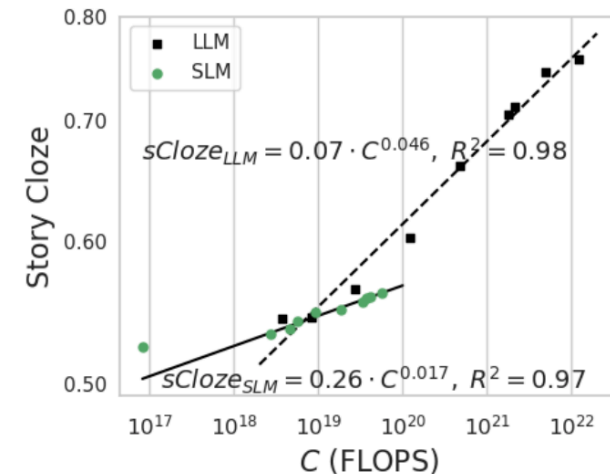
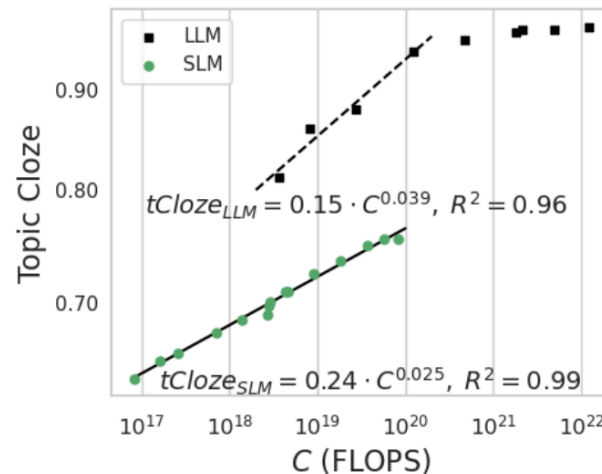
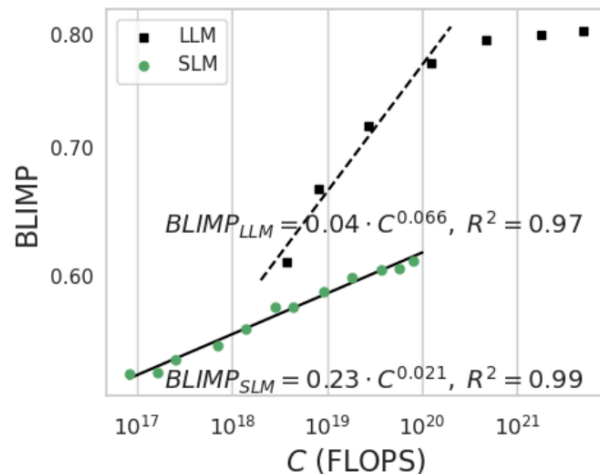


Prompt: *I really enjoy playing outside...*

Continuation: I really like to play outside because it forces me to really learn and really like, stay in my zone, I like that.

TEXTLESS-SLMs

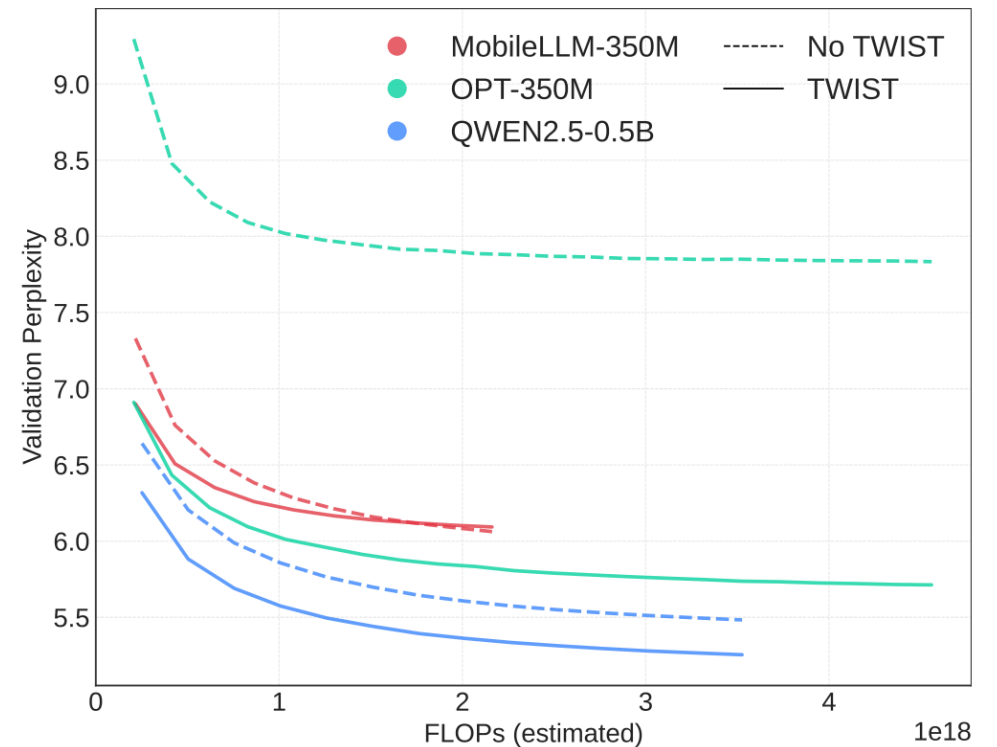
- Encouraging results!
- However, current SLM scaling laws show a somewhat pessimistic view.
 - Cuervo, Santiago, and Ricard Marxer. "Scaling properties of speech language models." *EMNLP* (2024).
- *tl;dr*: according to their study, we need 3x more data than text LMs!!!





TEXTLESS-SLMS

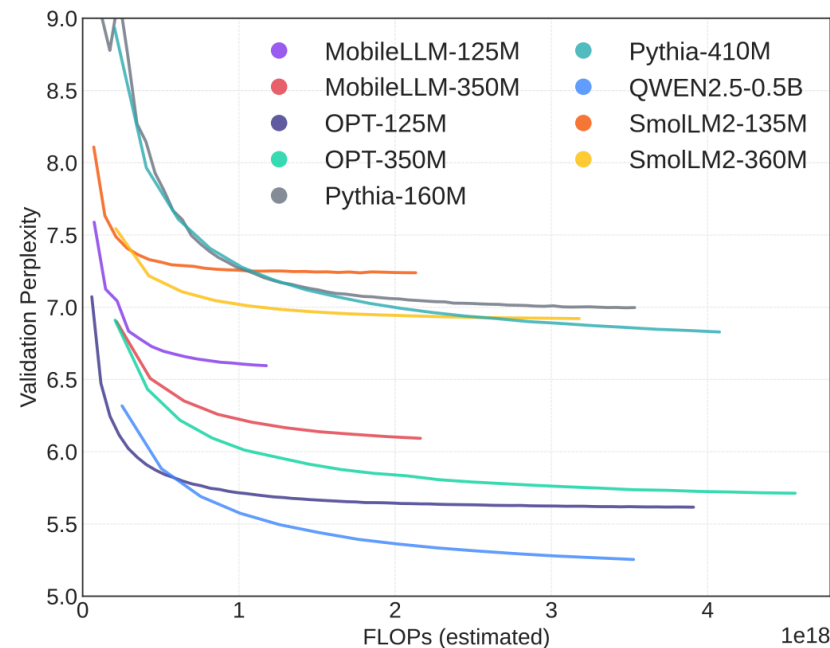
- But how does TWIST initialization effect these scaling laws?
- What about model architecture? Some architectures might benefit more / less?
- Should we expect to get better performance following such an approach?





TEXTLESS-SLMS

- SO TWIST is good! Lets deep dive into the effect of model initialization?
- Bigger models does not always perform better!





JOINT SPEECH-TEXT SLMS

- Will it be different when we incorporate text?
- How should we incorporate text?

(a) Mini-Omni

3	77	23	12	71	34	3	23
how	are	you	ϵ	ϵ	ϵ	ϵ	ϵ

(b) LLaMA-Omni

ϵ	3	77	23	ϵ	ϵ	12	71
how	ϵ	ϵ	ϵ	are	ϵ	ϵ	ϵ

(c) Moshi

3	77	23	12	71	34	3	23
how	ϵ	ϵ	are	ϵ	you	ϵ	ϵ

(d) SpiRit-LM

how	12	71	you	16	3	88	15
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SPEECH-TEXT INTERLEAVING

	Web Questions		Llama Questions		TriviaQA	
	S	S->T	S	S->T	S	S->T
Text LMs (text only)	-	32.3	-	75.0	-	56.4
Moshi	9.2	26.6	21.0	62.3	7.3	22.8
Zeng et al. w. syn. data	15.9	32.2	50.7	64.7	26.5	39.1

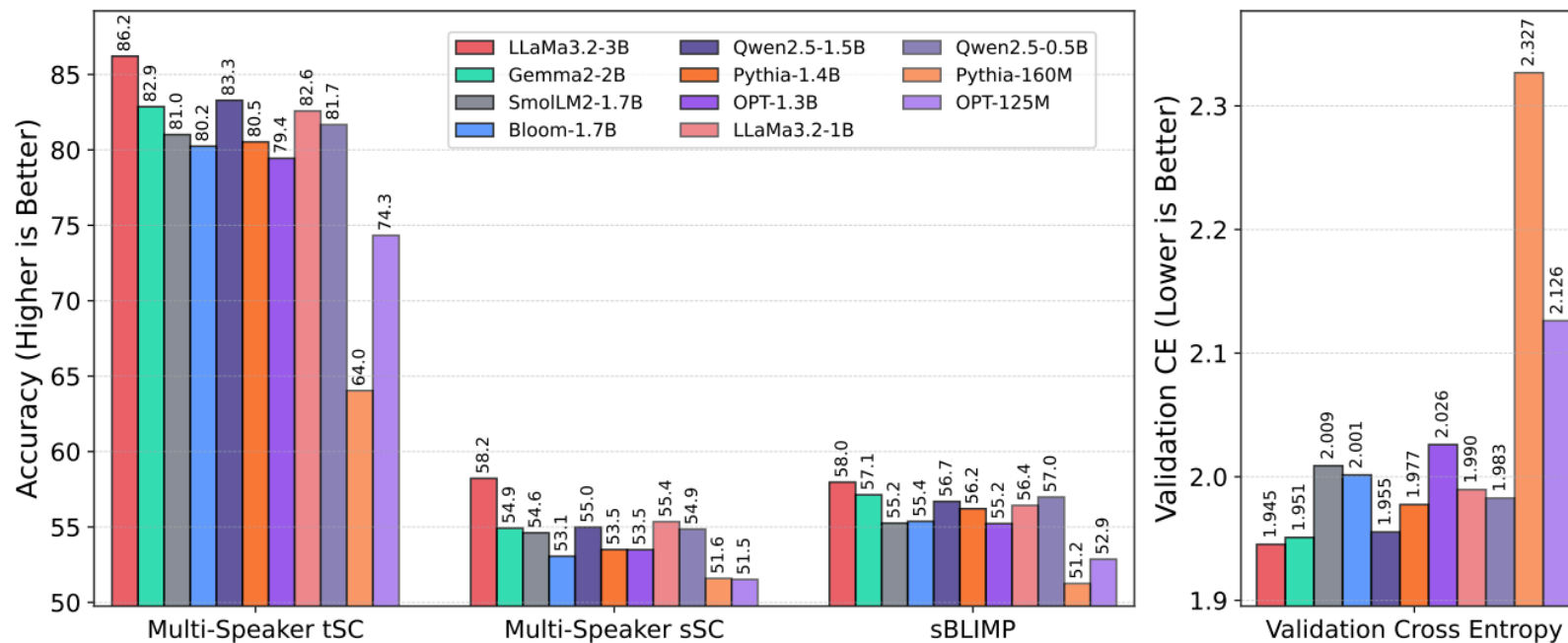


SPEECH-TEXT INTERLEAVING

Training Data		Metric					
Real	Syn.	sBLIMP↑	sSC↑	MS_sSC↑	tSC↑	MS_tSC↑	Val. CE↓
✓	✗	56.77	54.94	52.66	72.15	78.93	1.96129
✗	✓	52.98	61.30	54.84	81.24	72.35	3.44569
✓	✓	56.98	59.81	54.85	81.51	81.67	1.98267

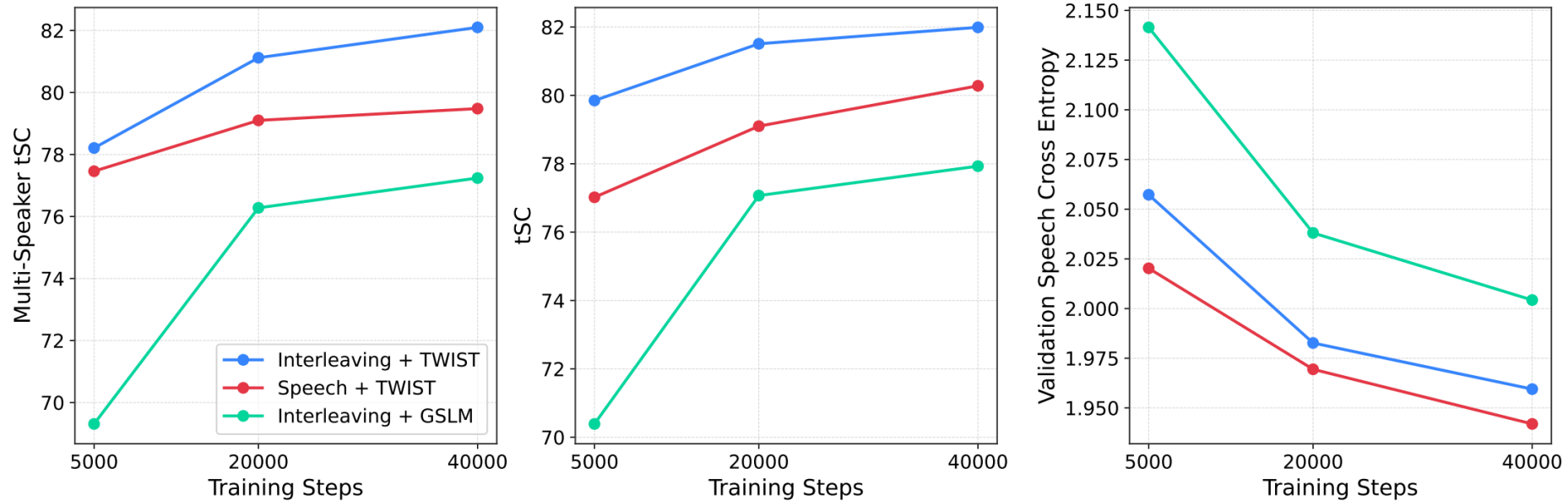
SPEECH-TEXT INTERLEAVING

- So interleaving is great, and synthetic data too
- What is the effect of model type / family under the interleaving setup



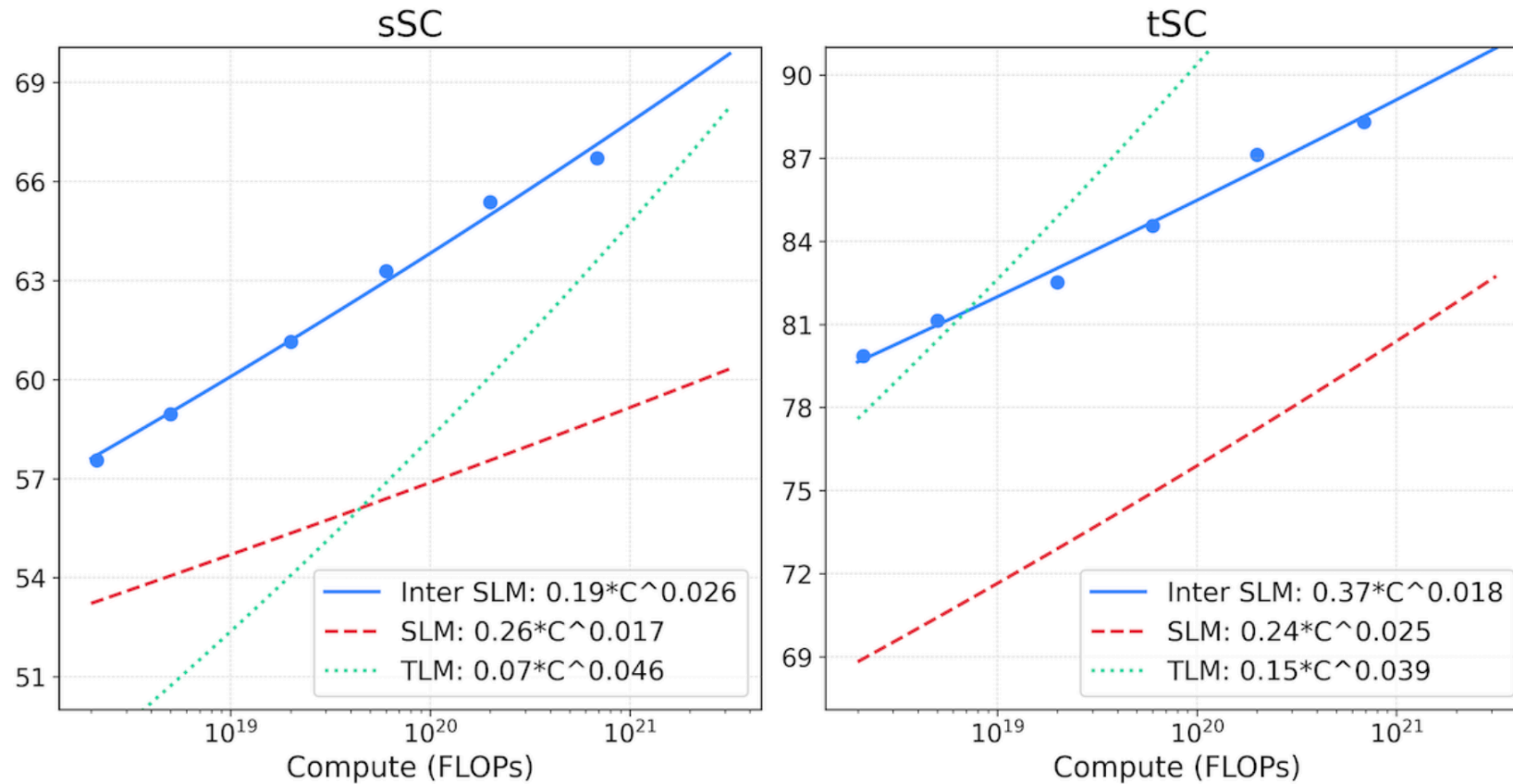
SPEECH-TEXT INTERLEAVING

- Do we need both TWIST and speech-text interleaving?



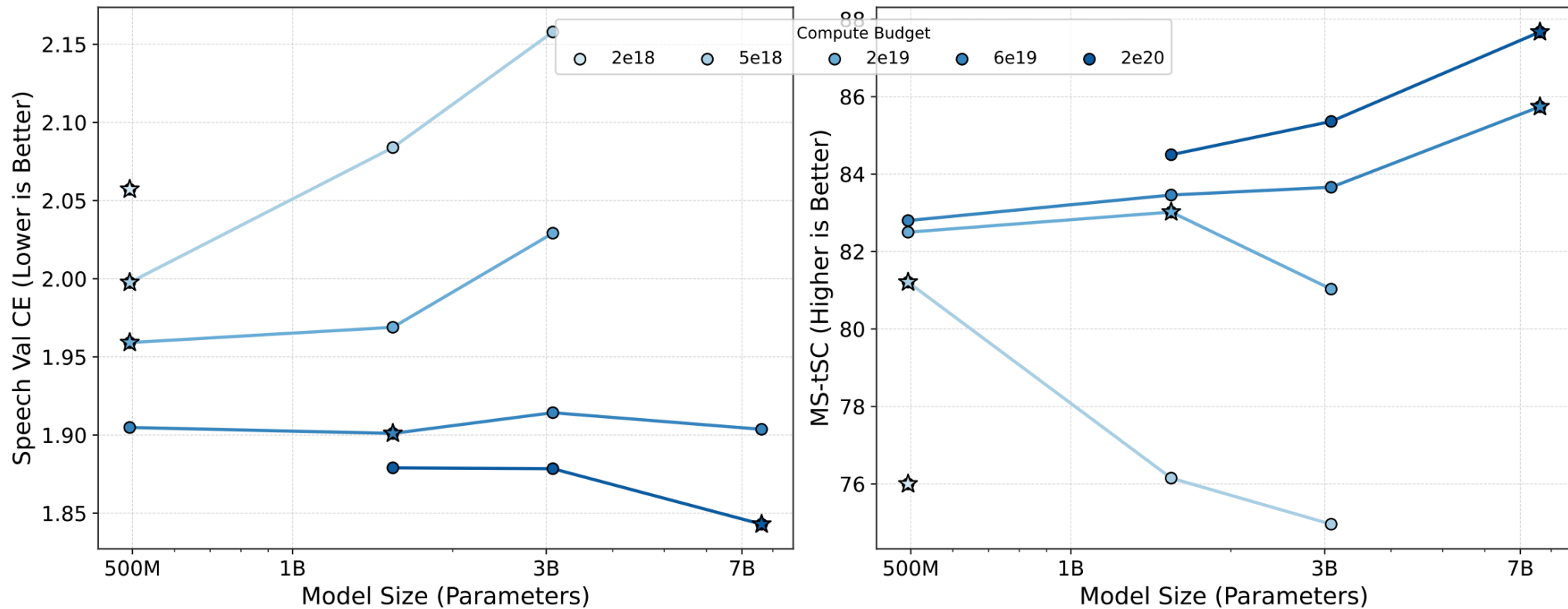


SPEECH-TEXT INTERLEAVING





SPEECH-TEXT INTERLEAVING



Maimon, Gallil, et al. "Scaling Analysis of Interleaved Speech-Text Language Models." *arXiv preprint arXiv:2504.02398* (2025).

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Prompt: *The capital of France is*

Continuation: is Paris, that's the city where we live. Paris is famous for it's culture and glory, and the culture is....

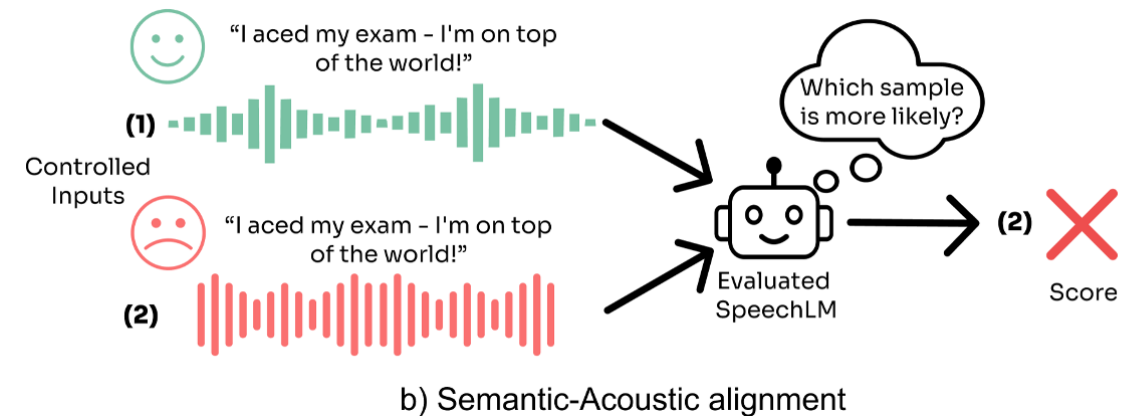
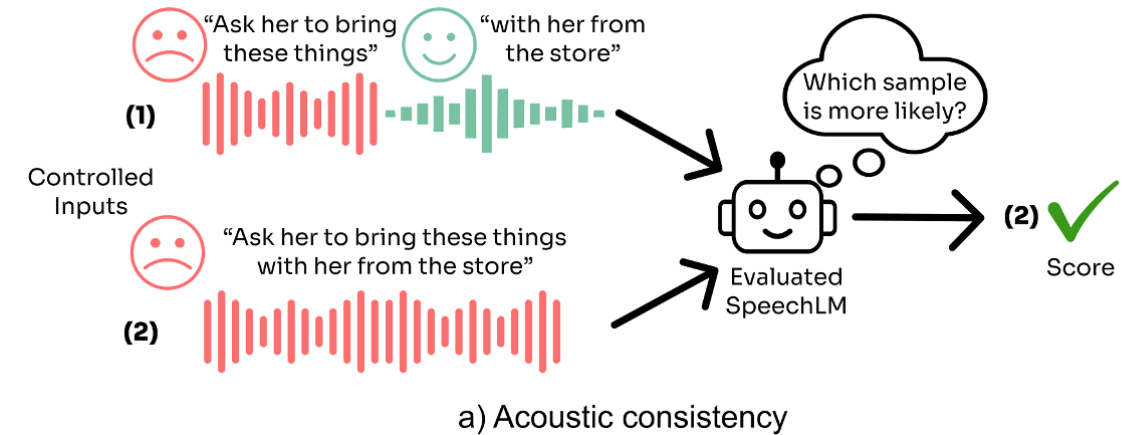


SPEECH-TEXT SLMS

- Great progress!
- But all evaluations are based on textual content :(
- Topline: ASR -> LLM -> TTS
- Can we design a benchmark / evaluation task that will leverage the richness of spoken data?
- Benchmark in which text based LLMs would perform at a chance level?
- *SALMON: Maimon, Gallil, Amit Roth, and Yossi Adi. "A suite for acoustic language model evaluation." ICASSP, 2025.*

SALMON

- Draw inspiration from early NLP benchmarks
- Create pairs of audio samples: positive and negative
- Negative has same content, but unlikely acoustics
- Ask SpeechLM “*which sample is more likely?*”
- Each sub-set of the metric checks one aspect

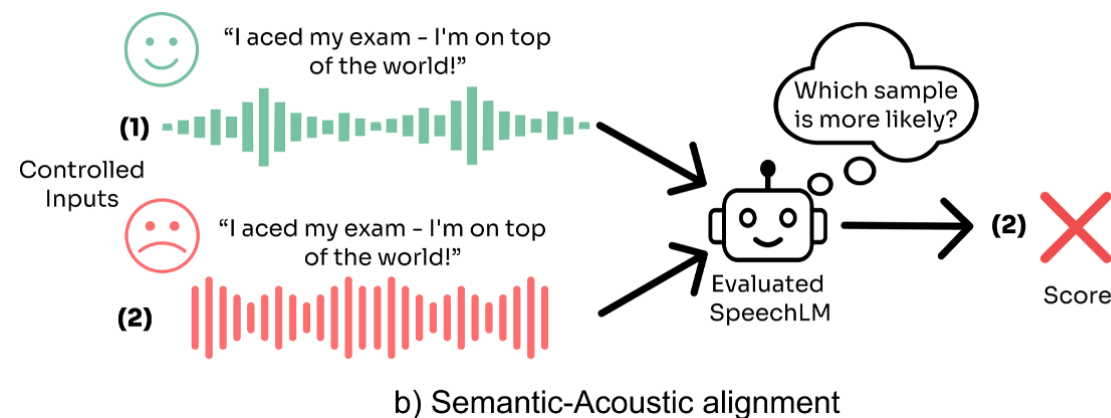
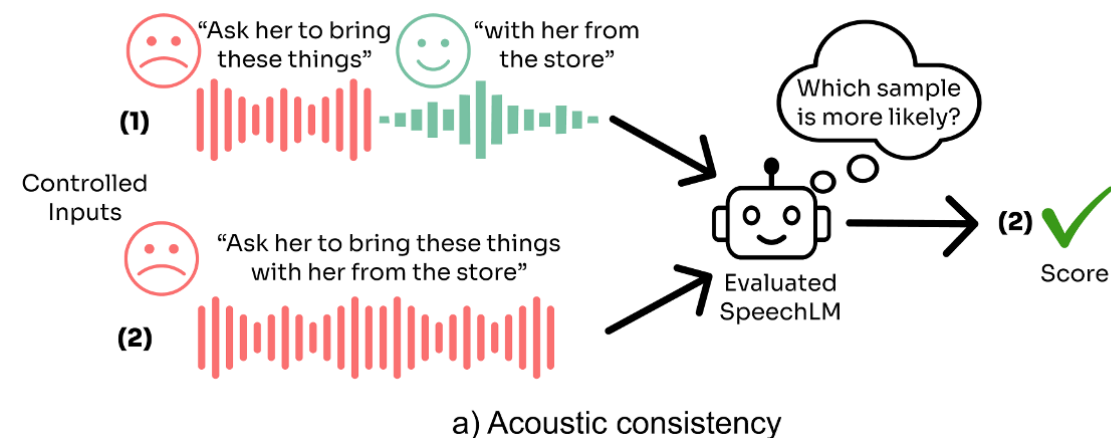


SALMON

- Speech data:
 - VCTK, LJ
 - Espresso
 - AzureTTS
- Background noise:
 - FSD50K
- RIR:
 - EchoThief
- GPT4 - generate sentences for alignment tasks



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SALMON - EXAMPLES

Speaker
Consistency

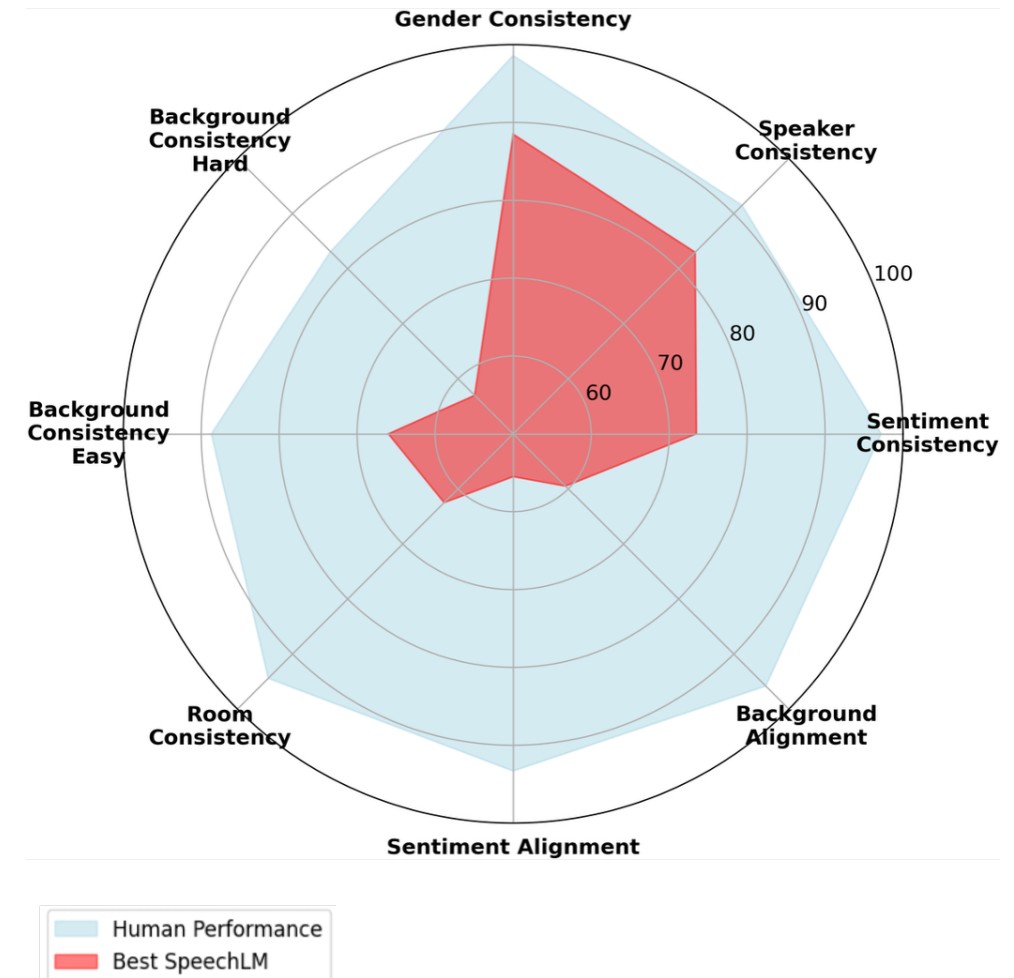
Emotion
Consistency

Background
Alignment



SALMON

- The task is trivial to humans
- Even expressive SpeechLMs struggle to detect basic inconsistencies
- In detecting mis-match between acoustic elements and content - results are random!





MULTI-MODAL SLMS

- Great progress!
- But all evaluations are based on textual content :(
- Topline: ASR -> LLM -> TTS
- Can we design a benchmark / evaluation task that will leverage the richness of spoken data?
- Benchmark in which text based LLMs would perform at a chance level?
- *SALMON: Maimon, Gallil, Amit Roth, and Yossi Adi. "A suite for acoustic language model evaluation." ICASSP, 2025.*
- ***Sentence Stress!***



SENTENCE STRESS

- Sentence stress refers to emphasis, placed on specific words within a spoken utterance to highlight or contrast an idea, or to introduce new information.

Stress Type	Description	Stressed speech (intention)
Contrastive	Demonstrates contrast with another option	" <i>I didn't take your book.</i> " (vs. someone else) " <i>I didn't take your book.</i> " (vs. something else)
Emphatic	Amplifies or diminishes the intensity of a concept.	" <i>They loved how you treated her dog.</i> " (You really exceeded their expectations)
New-Information	Marks a surprising or novel content in the discourse.	" <i>He's actually moving to New York.</i> " (surprising since its far from his current home)
Focus	General-purpose mechanism for highlighting key elements.	" <i>I enjoy the taste of espresso at sunrise.</i> " (It's about that particular time)

STRESS-TEST

- Yosha, Ido, et al. *StressTest*: Can YOUR Speech LM Handle the Stress?
- A **single-speaker** dataset (recorded by a professional actor) comprising 101 **manually curated** unique texts, each recorded with at least two distinct sentence stress pattern.

"You want to help me cook dinner tonight?"



SSD: What words did the speaker stress?

SSR: What is the underlying intention of the speaker?



SSD: The speaker stressed "you, me".

SSR: The other person might be a bad cook compared to the speaker.

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They never answer my calls

Highlighting that it absolutely never happened.

They never answer my calls

They might answer someone else's calls.



STRESS-TEST

- Model performance is measured in accuracy
- LLM-as-a-Judge
 - GPT-4o

$$\text{SSR}_{acc}(\mathcal{M}, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{(a,t,s,A,l) \in \mathcal{D}} \mathbb{I}\{\mathcal{J}(\mathcal{M}(a, \mathcal{P}(A))) = l\}.$$

- Where:
 - M - model, D - dataset, P - prompt, J - Judge
 - a - audio signal, A - possible set of answers, l - target answer



STRESS-TEST - AUDIO ONLY

Model	SSR
Qwen2Audio-7B-Instruct [Chu et al., 2024]	56.4
SALMONN [Tang et al., 2024]	56.8
LLaMA-Omni [Fang et al., 2025]	53.6
Phi-4-multimodal-instruct [Microsoft et al., 2025]	53.2
gpt-4o-audio [Hurst et al., 2024]	58.7
Human	96.1

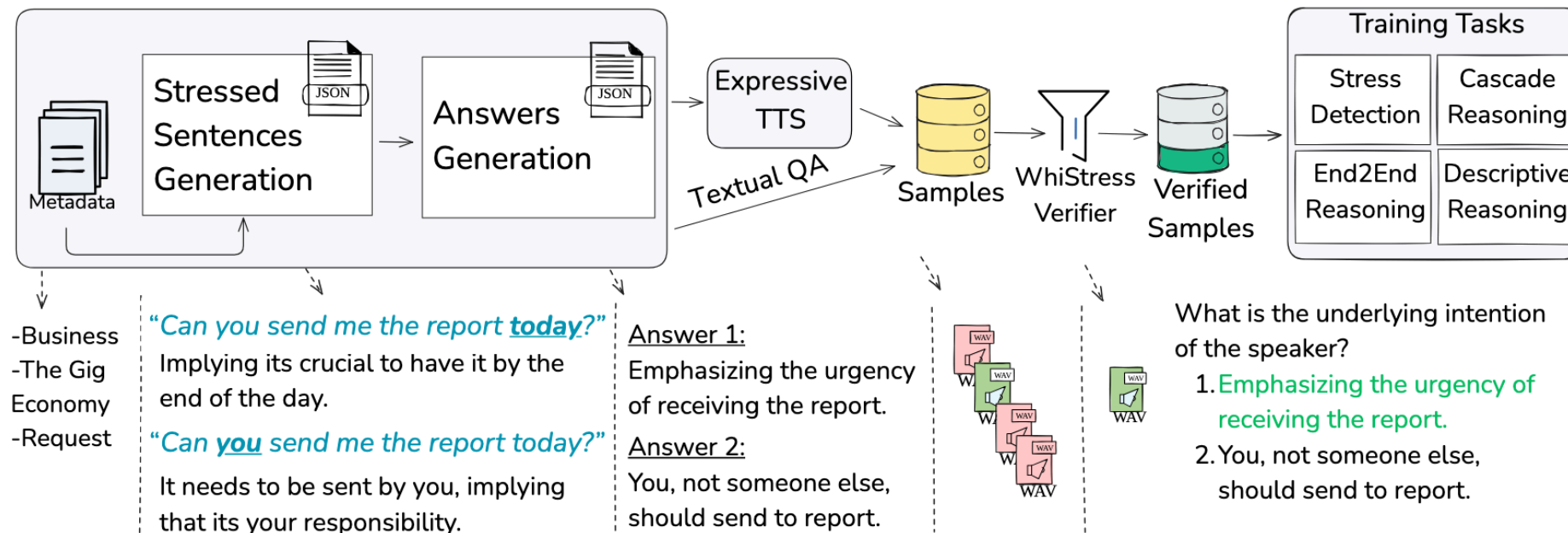


STRESS-TEST - TEXT W. STRESS

Model	SSR
<i>TextLM w. Oracle transcription + sentence stress</i>	
Llama-3.1-8B-Instruct [Grattafiori et al., 2024]	73.3
Qwen2-7B-Instruct [Yang et al., 2024a]	67.8
Qwen-7B-Chat [Bai and et al., 2023]	61.4
<i>Cascade models</i>	
Whiper+WhiStress → Llama-3.1-8B-Instruct	66.9
Whiper+WhiStress → Qwen2-7B-Instruct	63.7
Whiper+WhiStress → Qwen-7B-Chat	55.5

SYNTHETIC DATA GENERATION

- ~17k hours of synthetic data. ~4.5k hours were verified.
- 4 type types of tasks





STRESS-SLM (SSR)

- Fine-tune Qwen2Audio-7B-Instruct model using our data
 - Including rehearsal data
 - Curriculum learning

Model	SSR
Qwen2Audio-7B-Instruct [Chu et al., 2024]	56.4
SALMONN [Tang et al., 2024]	56.8
LLaMA-Omni [Fang et al., 2025]	53.6
Phi-4-multimodal-instruct [Microsoft et al., 2025]	53.2
gpt-4o-audio [Hurst et al., 2024]	58.7
StressSLM (ours)	81.6



STRESS-SLM (SSD)

Model	Precision	Recall	F1
gpt-4o-audio [Hurst et al., 2024]	33.1	52.1	40.5
SALMONN [Tang et al., 2024]	19.1	29.5	23.2
LLaMA-Omni [Fang et al., 2025]	24.1	47.6	32.0
Phi-4-multimodal-instruct [Microsoft et al., 2025]	19.9	32.8	24.7
Qwen2Audio-7B-Instruct [Chu et al., 2024]	24.6	46.2	32.1
StressSLM (ours)	89.6	83.3	86.4
WhiStress (verifier)	88.8	88.1	88.5



STRESS-SLM (SSD) - EXPRESSO

Model	Precision	Recall	F1
gpt-4o-audio [Hurst et al., 2024]	23.6	66.1	34.7
SALMONN [Tang et al., 2024]	13.2	45.5	20.5
LLaMA-Omni [Fang et al., 2025]	18.7	58.2	28.3
Phi-4-multimodal-instruct [Microsoft et al., 2025]	22.5	37.5	28.2
Qwen2Audio-7B-Instruct [Chu et al., 2024]	34.2	30.6	32.3
StressSLM (ours)	51.8	68.6	59.1
WhiStress (verifier)	57.3	86.3	68.9

Nguyen, Tu Anh, et al. "Expresso: A benchmark and analysis of discrete expressive speech resynthesis." *Interspeech* (2023).

Yosha, Iddo, et al. "WhiStress: Enriching Transcriptions with Sentence Stress Detection", *arXiv preprint arXiv:2505.19103* (2025).



STRESS-SLM

Verifier	Training Samples	SSD			SSR
		<i>Precision</i>	<i>Recall</i>	<i>F1</i>	<i>Acc.</i>
✓	~4K	87.3	76.3	81.4	79.3
✗	~17K	87.4	81.9	84.5	76.6
✗ → ✓	~17K → ~4K	88.3	83.7	85.9	78.4



STRESS-SLM

Model	ASR (WER)				SER
	<i>dev-clean</i>	<i>dev-other</i>	<i>test-clean</i>	<i>test-other</i>	<i>MELD</i>
Qwen2Audio-7B-Instruct	2.30	4.64	2.31	4.92	54.6
StressSLM (ours)	2.70	4.60	2.46	5.50	57.2



SUMMERY AND DISCUSSION

- Discussed a bit about SLMs
- Categorization and definitions
- SLMs scaling laws
- Joint Speech-Text Interleaving
- Multi-modality improves scaling properties!
- Benchmarking SLMs beyond spoken content



SUMMERY AND DISCUSSION

- What do we really want to get from these models?
 - An interface to textLLMs vs. universal speech processing systems
- What do we want SLMs to support?
 - ASR, TTS, reason over audio, source-separation, denoising, etc.
 - Each will affect the modeling and benchmark choices
- For instance, in textlessSLMs we should not expect to text on factual knowledge (e.g., QA)
- Goal should be to mimic a 3-5 years old, not a knowledge resource
- For speech-aware SLM, we should focus on speech properties! basically a speech interface to LLMs



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THANKS!

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